Question 1:

Using the support vectors identified in part a (ID = 1, 5, 6) we can derive the separating boundary and the geometric margin.

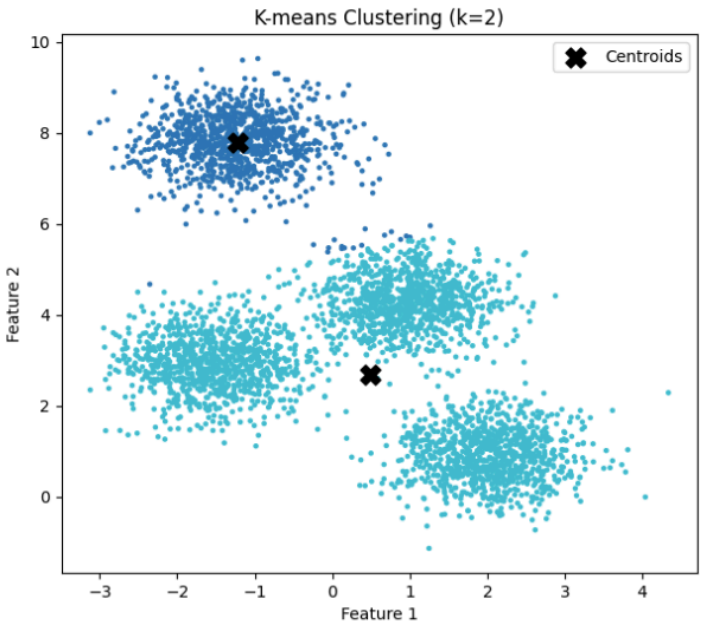
```
# w_1 x_1 + w_2 x_2 + b = 0
    # equation 1) w_1 + w_2 + b = 1
    # equation 2) w_1 + b = -1
    # equation 3) w_2 + b = -1

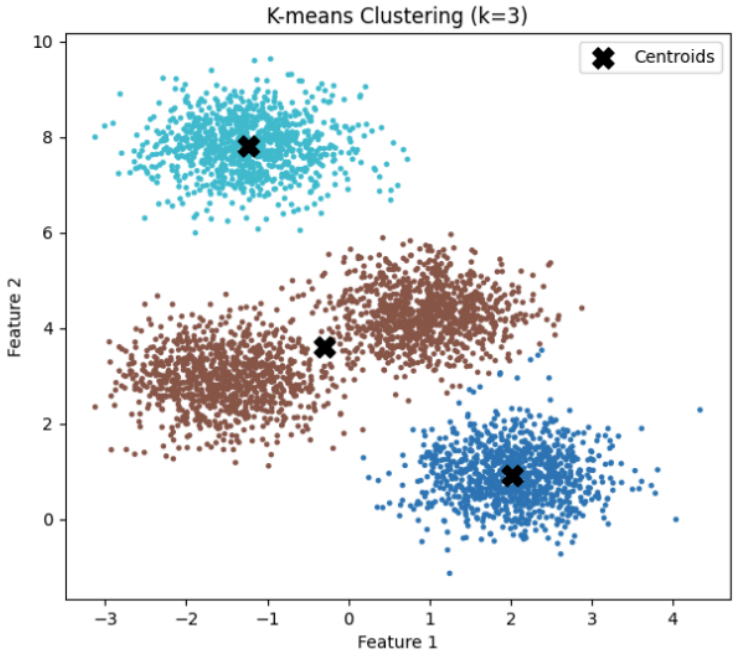
    # using eq 1 and 2 we get w_2 - 1 = 1 -> w_2 = 2
    # using eq 1 and 3 we get w_1 - 1 = 1 -> w_1 = 1
    # using eq 3 we get 2 + b = -1 -> b = -3
```

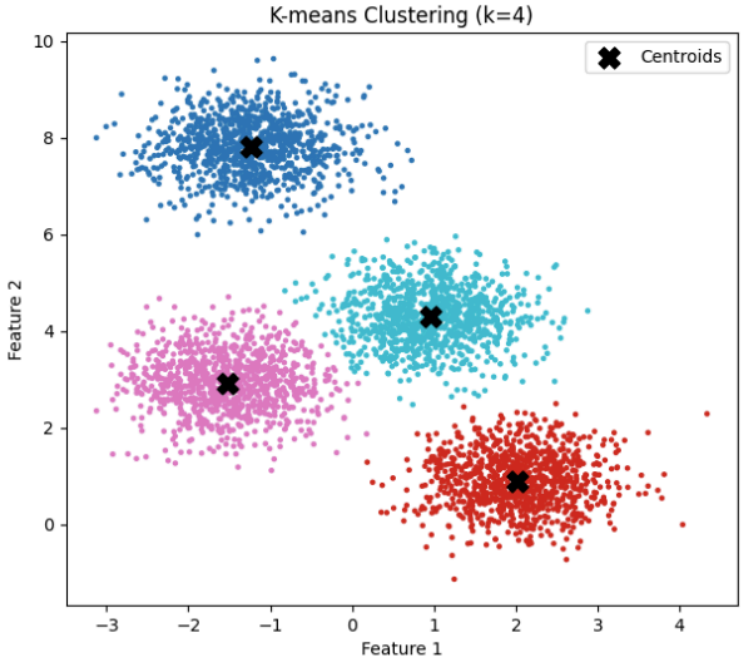
We start with the standard decision boundary equation w_1 x_1 + w_2 x_2 + b = 0 and will substitute our identified support vectors accordingly, seen below the starting equation. We can then solve the system of equations (3 unknowns, 3 equations) to get our weights and biases, seen above. Plugging our weights and bias back into the starting equation we get our separating boundary as: 2x_1 + 2x_2 -3 = 0

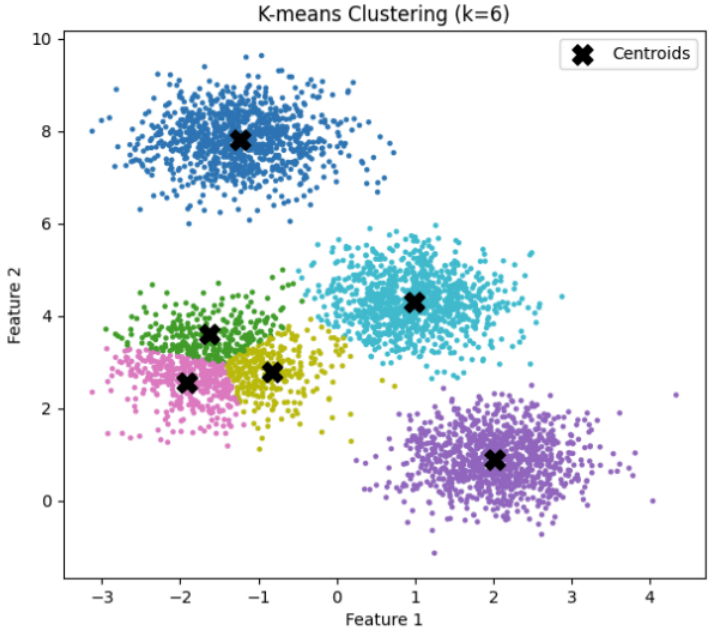
In order to get the geometric margin we find the distance from the closest point to the line. We plug into the equation to find that geometric_margin = 1 / (2^2 + 2^2) ^ (½) = 0.353

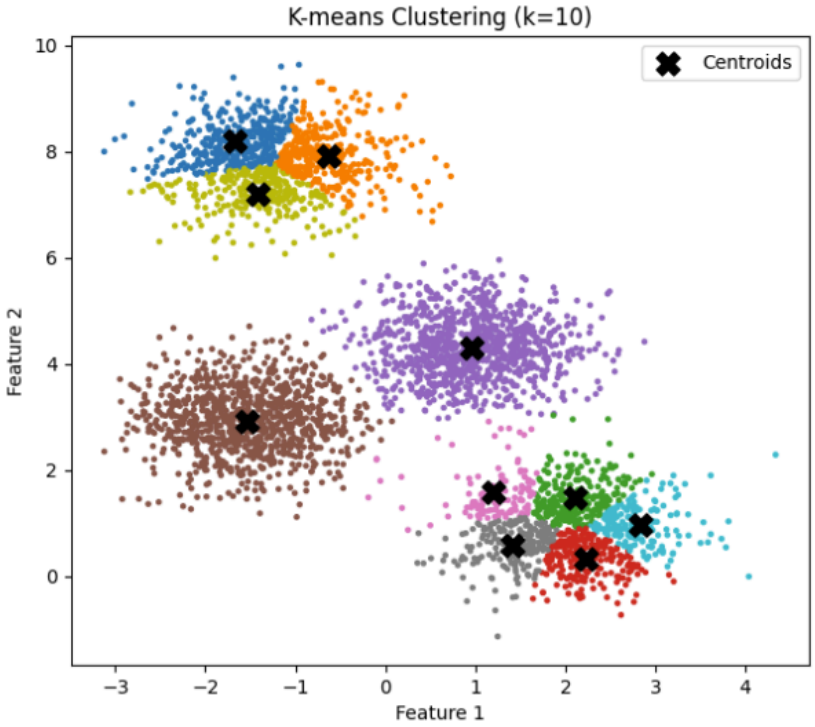
Question 2:




K-means Clustering (k=3)
Centroids
Feature 1
Feature 2
10
8
6
4
2
0
−3
−2
−1
0
1
2
3
4


K-means Clustering (k=4)
Centroids
Feature 1
Feature 2
10
8
6
4
2
0
−3
−2
−1
0
1
2
3
4





Above are the results from the K means clustering experiments. Results were gathered by calculating the euclidean distance from each point to each centroid and making the assignment accordingly. As we increase K, more clusters are created. We can see the clusters are very well

defined at lower K values, for example the 4 very obvious clusters in the data divide nicely into the 4 blobs seen in the plot above. As we increase K more clusters must be found, and we start to see the obvious blobs be divided into multiple clusters as a result of this.

Question 3:

The baseline mean accuracy for part a is 82.8%.
Using the KNN repair method I was able to get an accuracy of 86.44% across 20 trials.
The difference in results was 3.64%.

We can see a noticeable increase in accuracy over baseline (3.64% at already above 80% is quite good in my opinion). This tells us that using the nearest neighbors to correct against our introduced random noise helped the SVM learn a better decision boundary. This is likely because using neighbors in the decision can help outweigh the noise that was introduced into the data, and allow for correct classification despite this noise.